

Minimum-Peak Control of Fifth-Order Integrators via Exact Semidefinite Lifts

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Archive: ESM_1.zip | Version 2026-10-09

Access: [Versioned GitHub repository release](#). These materials are repository-hosted, not a journal-uploaded supplement. The journal name indicates the intended venue, not acceptance or publication.

Content description

Source code, numerical records, rational verification scripts and a reproducibility guide for the exact semidefinite formulation and its computational studies, including warnings, failed cases and boundary diagnostics.

Package identification and integrity

This PDF is the reading guide inside ESM_1.zip. The historical archive and metadata filenames are retained for compatibility. The machine-readable bundle-manifest.json carries the article, journal, author, affiliation and email for the archive and each listed file, together with its size and SHA256 checksum. Identification is stored separately to preserve the bytes of the frozen scientific files.

The manuscript PDF is at output/pdf/representability-review.pdf. Its source, templates and detailed claim-to-file guide are under research/paper/. The metadata record is research/paper/ONLINE_RESOURCE_1.json.

Quick replay without numerical libraries

Extract the complete ZIP to a new directory. Run from that directory:

```
python -B -S research/paper/review_checks.py
```

This verifies the inventory before and after checking source structure, reference metadata and exact fifth-order certificates. It uses only Python's standard library, does not call an optimizer and does not produce new solve timings.

Reading and verification guide

Where to find the evidence

Evidence	Directory or file (relative to the ZIP root)
Detailed instructions and claim map	research/paper/REPRODUCIBILITY.md
Four-moment prefix	research/moment_four_lift/
Weighted identity	research/schur_remainder_recursion/
Boundary witnesses and 21-block model	research/closed_five_lift/
Fifth-order synthesis and feedback	research/five_synthesis/; research/five_feedback/
Boundary and recovery diagnostics	research/theory_alignment/
Fourth-order comparisons	research/lift_value_benchmark/; research/peak_synthesis/
Supplementary symbolic checks	research/math_derivation_audit/20261002/

Extended replay

With the libraries listed in requirements-numerical.txt installed, run:

```
python -B research/paper/review_checks.py --numerical
```

This checks polynomial identities, matrix entries, boundary witnesses, stored studies and regression tests. Solver entry points are blocked during replay. Do not use -O or PYTHONOPTIMIZE: the historical checkers contain assertions. Libraries are not bundled, and the version list is not a transitive lockfile.

Interpretation and preservation

Expected fifth-order records include 30 valid certificates, accuracy targets met on 14/15 SDP and 12/15 dual tasks, and 16 feedback episodes with 192 slots. The SDP is slower on all 12 jointly accurate tasks and activates no plans under the recorded half-second deadline. Warnings, failures and the interrupted theory-alignment run remain available.

Replay checks finite records and algebra, not the universal analytic proofs, global novelty or fresh solve times. Use new output directories for new experiments; do not run creation or write flags on the distributed records. Checksums establish internal consistency, not independently timestamped provenance.